



Cyberster Blue Team Internship

Week 11 Report

Insider Threat Investigation: Confidential Data Leak at M57.biz

Prepared by: Muhammad Ubaid

Track: Blue Team Operations

Phase: Digital Forensics and Threat Intelligence

Submitted To: Sir Abdullah Zia (SOC Training Instructor)

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Core Competencies:

**Email Forensics & Header Analysis | LNK File Structural
Decomposition | Windows Prefetch Execution Profiling | Registry
Time-Key Correlation | Super Timeline Generation & Filtering |
Multi-Layer Evidential Synthesis**

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1. Overview

This week focused on a full DFIR investigation around a single confidential spreadsheet (m57biz.xls) belonging to Jean, CFO of [M57.biz](#). The file was leaked to a competitor, and Jean claims she only emailed it to the company President (Alison).

The investigation anchored on:

- Target file: m57plan.xls / m57biz.xls
- MD5: e23a4eb7f2562f53e88c9dca8b26a153
- Anchor time: 2008-07-20 01:28:03 GMT
- Window: 01:25:00 – 01:30:00 GMT

The case followed a "Start Big, Filter Down" approach: pull email evidence → confirm with host artefacts (LNK, Prefetch, Registry) → reconstruct second-by-second via a super timeline.

2. Methodology & Tools

Timezone

All analysis was conducted in GMT/UTC to match the threat intel window. Autopsy's data source timezone was set to UTC during ingestion.

Chain of Custody

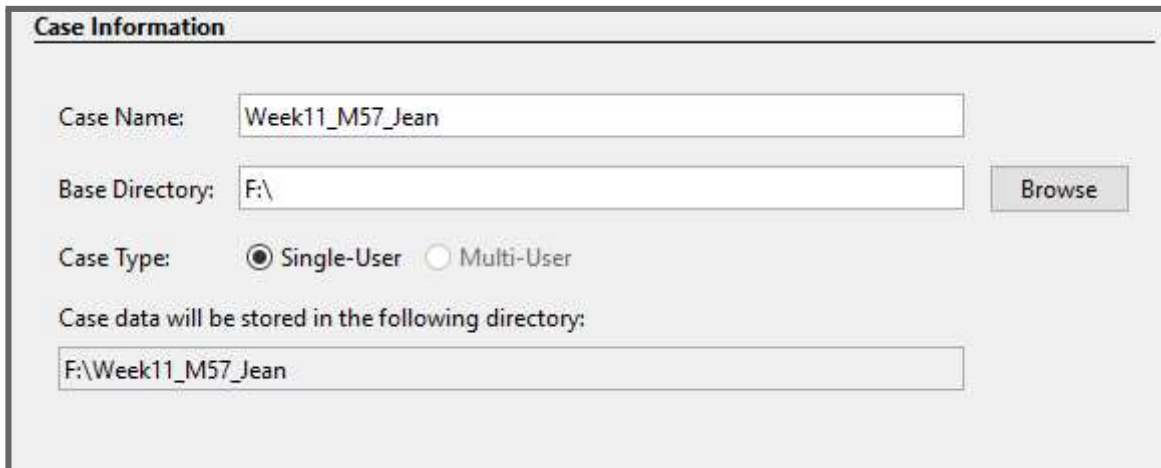
- Evidence files nps-2008-jean.E01 and .E02 were kept in a single read-only evidence folder.
- All extracted artefacts (LNK, Prefetch, hives, PST, timeline) were saved to a separate working folder.
- MD5 verification was performed in Autopsy and matched the original image hash.

Tools Used

Tool	Purpose
Autopsy	Main forensic platform
LECmd	Parsing .lnk shortcuts
PECmd	Parsing Prefetch files
Registry Explorer	NTUSER.DAT analysis
Microsoft Excel	Reviewing exported CSVs
Windows CMD	Running CLI tools

Case Setup

- Case Name: Week11_M57_Jean
- Type: Single-User
- Working Dir: F:\Week11\
- Data Source: nps-2008-jean.E01 (split image auto-detected .E02)
- Timezone: GMT/UTC
- Modules Enabled: Recent Activity, Hash Lookup, Keyword Search, Embedded File Extractor, Email Parser , File Type ID, Interesting Files, Encryption Detection



Case Information

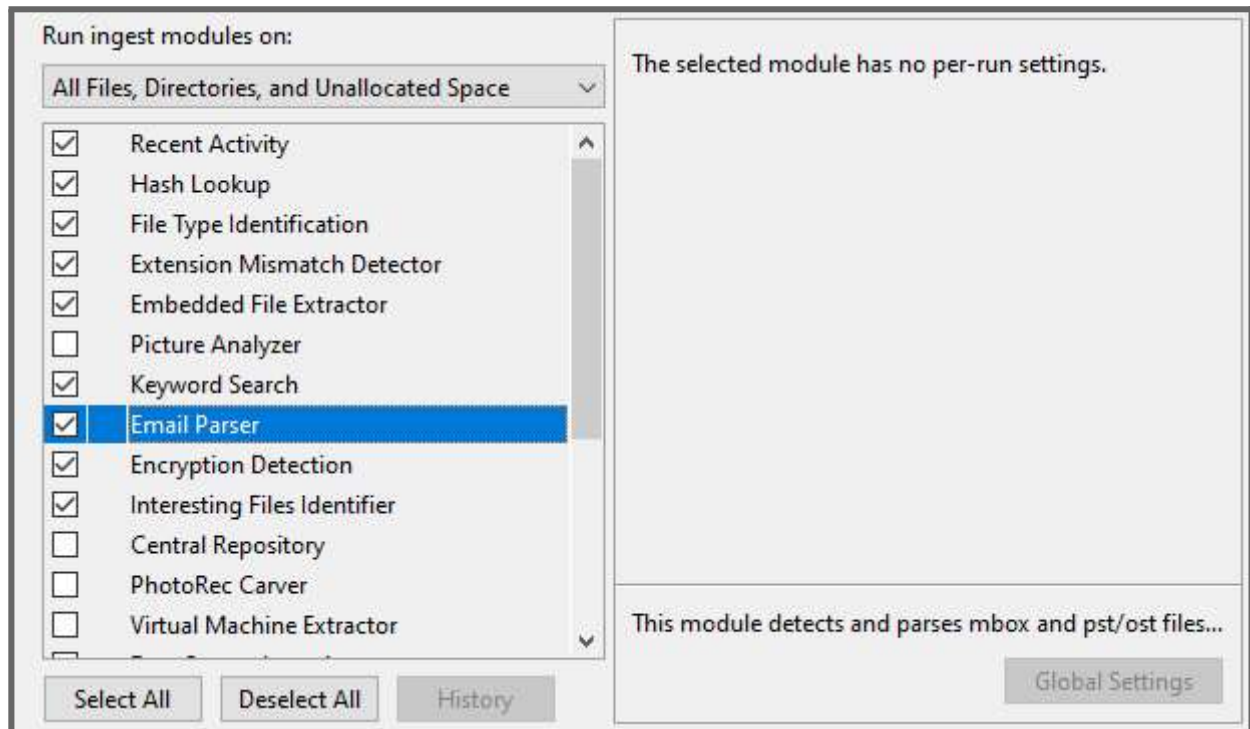
Case Name:

Base Directory:

Case Type: ☒ Single-User ☐ Multi-User

Case data will be stored in the following directory:

Ingest module configuration with Email Parser highlighted



Ingest module configuration with Email Parser highlighted



MD5 hash verification result confirming image integrity

3. Part A — Email Extraction & Triage (Days 74–75)

Objective

Find Jean's mail archive, locate the outbound email containing the spreadsheet, and check whether the displayed recipient matches the real underlying address.

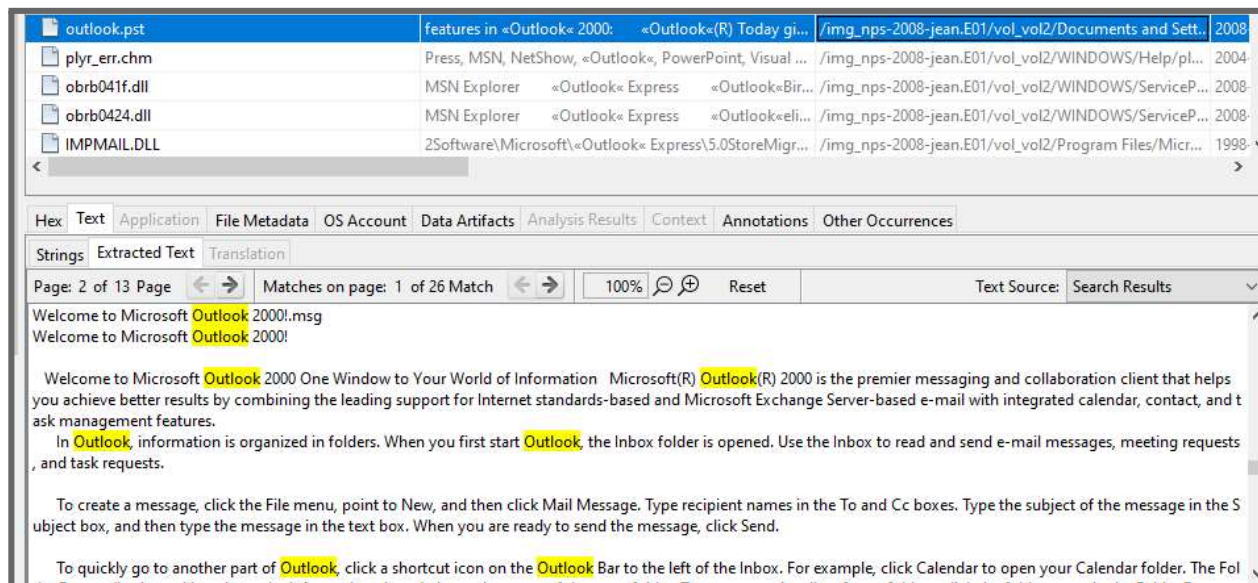
3.1 Email Client & Archive Located

Autopsy showed Jean was using Microsoft Outlook .

outlook.pst Metadata

Field	Value
Path	...\Microsoft\Outlook\outlook.pst
Size	2,326,528 bytes
Status	Allocated
Created	2008-07-06 07:38:42 UTC
Modified	2008-07-21 01:17:54 UTC
MIME	application/vnd.ms-outlook-pst
MD5	8c862a8c7ad8b7aff1df4d44fbf1fe95

The PST was last modified one day after the leak — consistent with Outlook updating after the send.



3.2 Target Email Identified

Autopsy's Email Parser produced 261 email records . The target email was found at result 261:

Field	Value
From	Jean User <jean@m57.biz>
Subject	RE: Please send me the information now
Sent	2008-07-20 01:28:00 UTC
Body	"I've attached the information that you have requested to this email message."

The body is suspiciously short — no verification, no questions.

The screenshot shows the Autopsy Email Parser interface. At the top, a list of email records is visible, with the selected record (Result 261) highlighted in blue. The record details are as follows:

Field	Value
From	Jean User <jean@m57.biz>
To	alison@m57.biz
CC	
Subject	RE: Please send me the information now

Below the headers, the email body is displayed in a text format. The body text is:

I'm sorry to bother you, but I really need that information now --- this VC guy is being very insistent. Can you please reply to this email with the information I requested --- the names, salaries, and social security numbers (SSNs) of all our current employees and intended hires?

Thanks.

Alison

3.3 The Critical Header Discrepancy

Inside the quoted original message at the bottom of Jean's reply, the real sender address was exposed:

Header Field	Displayed Value	Real Value
From (Display)	alison@m57.biz	—
From (Actual)	—	tuckgorge@gmail.com
Sent	2008-07-20 02:23 (BST)	5 min 37 sec before Jean's reply
To	jean@m57.biz	Confirmed
Subject	"Please send me the information now"	Social engineering pressure
This is the pivot of the case. The display name alison@m57.biz was spoofed — the actual delivery address was an external Gmail account.		

```
-----Original Message-----
From: alison@m57.biz [mailto:tuckgorge@gmail.com]
Sent: Sunday, July 20, 2008 2:23 AM
To: jean@m57.biz
Subject: Please send me the information now
```

Raw email headers showing display name alison@m57.biz next to real address tuckgorge@gmail.com

3.4 Attachment Analysis — m57biz.xls

Attachment details from the email:

- Size: 291,840 bytes (~285 KB)
- MIME: application/vnd.ms-excel
- Location: /Documents and Settings/Jean/Desktop/m57biz.xls

File on disk metadata:

| Field | Value |

|-----|-----|

| Path | Desktop/m57biz.xls |

| Size | 291,840 bytes |

| Created | 2008-07-20 01:28:03 UTC |

| Modified | 2008-07-20 01:28:03 UTC |

| Accessed | 2008-07-20 01:28:03 UTC |

| Changed | 2008-07-20 01:28:04 UTC |

| MD5 | e23a4eb7f2562f53e88c9dca8b26a153 ☒ (matches brief) |

| Internal ID | 4014 |

All four MAC timestamps cluster within 1 second of the anchor — strong evidence the file was attached and sent immediately.

M57.biz company				
Name	Position	Salary	SSN (for background check)	
Alison Smith	President	\$140,000	103-44-3134	
Jean Jones	CFO	\$120,000	432-34-6432	
Programmers:				
Bob Blackman	Apps 1	90,000	493-46-3329	
Carol Canfred	Apps 2	110,000	894-33-4560	
Dave Daubert	Q&A	67,000	331-95-1020	
Emmy Arlington	Entry Level	57,000	404-98-4079	
Marketing				
Gina Tangers	Creative 1	80,000	980-97-3311	
Harris Jenkins	G & C	105,000	887-33-5532	
BizDev				
Indy Counterchng	Outreach	240,000	123-45-6789	
Annual Salaries			\$1,009,000	

Metadata

Name: /img_nps-2008-jean.E01/vol_vol2/Documents and Settings/Jean/Desktop/m57biz.xls
Type: File System
MIME Type: application/vnd.ms-excel
Size: 291840
File Name Allocation: Allocated
Metadata Allocation: Allocated
Modified: 2008-07-20 07:28:03 PKST
Accessed: 2008-07-20 07:28:03 PKST
Created: 2008-07-20 07:28:03 PKST
Changed: 2008-07-20 07:28:04 PKST
MD5: e23a4eb7f2562f53e88c9dca8b26a153
SHA-256: 34456b5f714dc9d8dd23c742d54c3f5f582ecb042bc1c4d3042b88203863779f
Hash Lookup Results: UNKNOWN
Internal ID: 4014

Autopsy File Metadata tab for m57biz.xls showing MD5 match

3.5 Spreadsheet Contents (What Was Leaked)

The extracted text revealed a sheet titled "[M57.biz](#) company" containing the full employee directory with names, positions, salaries, and SSNs.

Sensitive data exposed included: - 9 employees (Executive, Programmers, Marketing, BizDev)

- Annual Salaries: **
- 1,009,000**--Benefits(30
- 1,009,000**--Benefits(30302,700
- Total Comp: **
- 1,311,700**--MonthlyBurn:
- 1,311,700**--MonthlyBurn:109,308.33
- Every employee's SSN

m57biz.xls Sheet1					
M57.biz company					
Name		Position	Salary	SSN (for background check)	
Alison	Smith	President	\$140,000	103-44-3134	
Jean	Jones	CFO	\$120,000	432-34-6432	
Programmers:					
Bob	Blackman	Apps 1	90,000	493-46-3329	
Carol	Canfred	Apps 2	110,000	894-33-4560	
Dave	Daubert	Q&A	67,000	331-95-1020	
Emmy	Arlington	Entry Level	57,000	404-98-4079	
Marketing					
Gina	Tangers	Creative 1	80,000	980-97-3311	
Harris	Jenkins	G & C	105,000	887-33-5532	
BizDev					
Indy	Counterching	Outreach	240,000	123-45-6789	
Annual Salaries				\$1,009,000	

Autopsy Extracted Text tab showing m57biz.xls contents

Part A — Initial Conclusion

The email evidence alone proves the file was not sent to the President. The real recipient address tuckgorge@gmail.com is an external Gmail account spoofed to look like an internal [m57.biz](#) contact. This is the signature of a spear phishing / social engineering attack .

4. Part B — Host Artefact Deep Dive (Days 76–78)

Objective

Corroborate the email evidence using three independent OS artefact categories: LNK files, Prefetch, and Registry .

4.1 LNK File Analysis

Two LNK files referencing m57biz.xls were found:

LNK Location	Size	Created (UTC)	MD5
Jean\Recent\m57biz.lnk	468 B	2008-07-20 01:28:04	267c4ad8a74c278fa0d5013342b43b64
Office\Recent\m57biz.LNK	408 B	2008-07-20 01:27:42	e0250a439c4b606dbb982c2902ff672c

The 22-second gap between the two represents the time Excel had the file open before the OS finalized the Windows-level shortcut.

Name	Keyword Preview
Unalloc_56144_6793707008_8345808384	Googlem57biz.xls«m57biz.lnk««m57biz.lnk«MRUListEx...
m57biz.LNK	«m57biz.lnk«
_REGISTRY_USER_NTUSER_S-1-5-21-484763869-7968	Googlem57biz.xls«m57biz.lnk««m57biz.lnk«MRUListEx...
m57biz.lnk	«m57biz.lnk«
SMFT	/body> </html>FILE0«m57biz.lnk« m57biz.xls m57biz.xls

Autopsy view of both LNK files in their respective Recent folders

Recent Documents Artifact

Autopsy's Recent Documents view returned 9 results. The m57biz.LNK entry showed:

- Target Path: C:\Documents and Settings\Jean\Desktop\m57biz.xls
- Date Accessed: 2008-07-20 01:27:42 UTC
- Path ID: 4014 ← matches Internal ID from Part A (cross-artifact link confirmed)

Source Name	S	C	O	Path	Date Accessed	Data Source
Desktop.LNK				C:\Documents and Settings\Jean\Desktop	2008-07-20 07:28:04 PKST	nps-2008-jean.E01
m57biz.LNK				C:\Documents and Settings\Jean\Desktop\m57biz.xls	2008-07-20 07:27:42 PKST	nps-2008-jean.E01
My Pictures.LNK				C:\Documents and Settings\Jean\My Documents\My ...	2008-07-06 13:50:42 PKST	nps-2008-jean.E01
m57biz.lnk				C:\Documents and Settings\Jean\Desktop\m57biz.xls	2008-07-20 07:28:04 PKST	nps-2008-jean.E01
tag-cloud.lnk				C:\Documents and Settings\Jean\Desktop\tag-cloud.j...	2008-07-12 00:00:37 PKST	nps-2008-jean.E01

Hex	Text	Application	Source File Metadata	OS Account	Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Result: 1 of 1 Result < >									
Type	Value								
Path	C:\Documents and Settings\Jean\Desktop\m57biz.xls								
Path ID	4014								
Date Accessed	2008-07-20 07:27:42 PKST								
Source File Path	/img_nps-2008-jean.E01/vol_vol2/Documents and Settings/Jean/Application Data/Microsoft/Office/Recent/m57biz.LNK								
Artifact ID	-9223372036854775796								

Autopsy Recent Documents tab showing m57biz.LNK with Path ID 4014

LECmd Output

Parsed with LECmd v2026.5.0 :

Field	Value
Target Created/Modified/Accessed	2008-07-20 01:28:03 UTC
File Size	291,840 bytes
Relative Path	..\Desktop\m57biz.xls
Drive Type	Fixed (Hard Drive)
Volume Serial	7A4FC21F
Machine ID	jean-13bf038a3
MAC Address	00:0c:29:8d:35:2b (VMware)
Tracker Created	2008-07-20 01:26:17 UTC

The Machine ID locks the activity to Jean's specific machine. The Tracker DB started 1m 25s before the file was opened, proving the tracking wasn't injected after the fact.

```
C:\Tools\LECmd.exe -f "D:\Week11\LNK\m57biz.lnk" --csv "D:\Week11\LNK"
LECmd version 2026.5.0

Author: Eric Zimmerman (ezriczimmerman@gmail.com)
https://github.com/EricZimmerman/LECmd

Command line: -f D:\Week11\LNK\m57biz.lnk --csv D:\Week11\LNK

Warning: Administrator privileges not found!

Processing D:\Week11\LNK\m57biz.lnk
Source file: D:\Week11\LNK\m57biz.lnk
Source created: 2026-05-17 18:18:58
Source modified: 2026-05-17 18:18:58
Source accessed: 2026-05-17 18:21:51

--- Header ---
Target created: 2008-07-20 01:28:03
Target modified: 2008-07-20 01:28:03
Target accessed: 2008-07-20 01:28:03

File size (bytes): 291,840
Flags: HasTargetList, HasInitInfo, HasRelativePath, HasWorkingDir, IsIconic
File attributes: FileAttributesArchive
Icon index: 0
Show window: ShowNormal (Activates and displays the window. The window is restored to its original size and position if the window is minimized.)

Relative Path: ..\Desktop\m57biz.xls
Working Directory: C:\Documents and Settings\Jean\Desktop

--- Link Information ---
Flags: VolumeIsRooted, HasPath

== Volume Information
Drive type: Fixed storage media (Hard drive)
Serial number: 7A4FC21F
Label: (No label)
Local path: C:\Documents and Settings\Jean\Desktop\m57biz.xls

--- Target ID Information (Format: Type ==> Value) ---
```

LECmd CLI output showing parsed LNK header and machine ID



LECmd CSV export opened in Excel showing all parsed fields

4.2 Prefetch Analysis

File: EXCEL.EXE-1C75F8D6.pf (30,510 bytes)

Field	Value
Created	2008-07-17 14:48:33 UTC
Modified / Last Run	2008-07-20 01:27:50 UTC (last update)
Last Execution (PECmd)	2008-07-20 01:27:40 UTC
Run Count	2
Volume Serial	744FC21F  matches LNK
Directories Referenced	31
File References	91

Excel was launched only twice on this system — once on July 17 (first install/use) and again on July 20 (the incident).

EXCEL.EXE-1C75F8D6.pf
1
2008-07-20 07:27:50 PKST
2008-07-20 07:27:50 PKST
2008-07-20 07:27:50 PKST

Hex
Text
Application
File Metadata
OS Account
Data Artifacts
Analysis Results
Context
Annotations
Other Occurrences

Metadata
Name: /img_nps-2008-jean.E01/vol_vol2/WINDOWS/Prefetch/EXCEL.EXE-1C75F8D6.pf
Type: File System
MIME Type: application/octet-stream
Size: 30510
File Name Allocation: Allocated
Metadata Allocation: Allocated
Modified: 2008-07-20 07:27:50 PKST
Accessed: 2008-07-20 07:27:50 PKST
Created: 2008-07-17 20:48:33 PKST
Changed: 2008-07-20 07:27:50 PKST
MD5: c7172b8375adcd5c2797237be5657be8
SHA-256: 3477d1adf9b2c86d22d4919b52d63481143ae09cac2c84484e3041bd457e25d6
Hash Lookup Results: UNKNOWN
Internal ID: 35001

Autopsy file metadata of [EXCEL.EXE-1C75F8D6.pf](#)


```

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/PECmd

Command line: -f D:\Week11\Prefetch\EXCEL.EXE-1C75F8D6.pf

Warning: Administrator privileges not found!

Keywords: temp, tmp

Processing D:\Week11\Prefetch\EXCEL.EXE-1C75F8D6.pf

Created on: 2020-05-17 18:31:47
Modified on: 2020-05-17 18:33:47
Last accessed on: 2020-05-17 18:33:58

Executable name: EXCEL.EXE
Hash: 3C75F8D6
File size (bytes): 30,538
Version: Windows XP or Windows Server 2003

Run count: 2
Last run: 2020-07-20 01:27:48

Volume information:

#1: Name: \DEVICE\HARDDISKVOLUME1 Serial: 794FC21F Created: 2008-05-15 23:18:42 Directories: 21 File references: 61

Directories referenced: 11

#1: \DEVICE\HARDDISKVOLUME1\
#2: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\
#3: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\ALL USERS\
#4: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\ALL USERS\APPLICATION DATA\
#5: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\ALL USERS\DOCUMENTS\
#6: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\ALL USERS\DOCUMENTS\MY MUSIC\

```

PECmd output showing Last Run timestamp and Run Count = 2] Critical files accessed during Excel session (PECmd flagged Keyword: True):

```

| # | File Accessed |
|---|-----|
| 38 | \DOCUME~1\JEAN\LOCALS~1\TEMP\M57BIZ.XLS |
| 40 | \DOCUME~1\JEAN\LOCALS~1\TEMP\~DFF9E0.TMP |
| 57 | \DOCUME~1\JEAN\APPLICATION DATA\MICROSOFT\OFFICE\RECENT\TEMP.LNK |

```

The temp copy and ~DF*.TMP working file confirm the spreadsheet was actively handled, not just listed.

```

#1: \DEVICE\HARDDISKVOLUME1\WINDOWS\SYSTEM32\WIN32K.SYS
#2: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\JEAN\APPLICATION DATA\DESKTOP.INI
#3: \DEVICE\HARDDISKVOLUME1\WINDOWS\SYSTEM32\LINKINFO.DLL
#4: \DEVICE\HARDDISKVOLUME1\WINDOWS\SYSTEM32\NTSHRUI.DLL
#5: \DEVICE\HARDDISKVOLUME1\WINDOWS\SYSTEM32\ATL.DLL
#6: \DEVICE\HARDDISKVOLUME1\WINDOWS\SYSTEM32\NETAPI32.DLL
#7: \DEVICE\HARDDISKVOLUME1\WINDOWS\SYSTEM32\USERENV.DLL
#8: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\JEAN\START MENU\DESKTOP.INI
#9: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\ALL USERS\START MENU\DESKTOP.INI
#10: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\ALL USERS\APPLICATION DATA\DESKTOP.INI
#11: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\JEAN\MY DOCUMENTS\DESKTOP.INI
#12: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\JEAN\MY DOCUMENTS\MY PICTURES\DESKTOP.INI
#13: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\ALL USERS\DOCUMENTS\DESKTOP.INI
#14: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\ALL USERS\DOCUMENTS\MY PICTURES\DESKTOP.INI
#15: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\ALL USERS\DOCUMENTS\MY MUSIC\DESKTOP.INI
#16: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\ALL USERS\DOCUMENTS\MY VIDEOS\DESKTOP.INI
#17: \DEVICE\HARDDISKVOLUME1\DOCUMENTS AND SETTINGS\JEAN\APPLICATION DATA\MICROSOFT\OFFICE\RECENT\TEMP.LNK (Keyword: True)
#18: \DEVICE\HARDDISKVOLUME1\PROGRAM FILES\MICROSOFT OFFICE\OFFICE\EXCEL.DIP

----- Processed D:\Week11\Prefetch\EXCEL.EXE-1C75F8D6.pf in 0.07566470 seconds -----

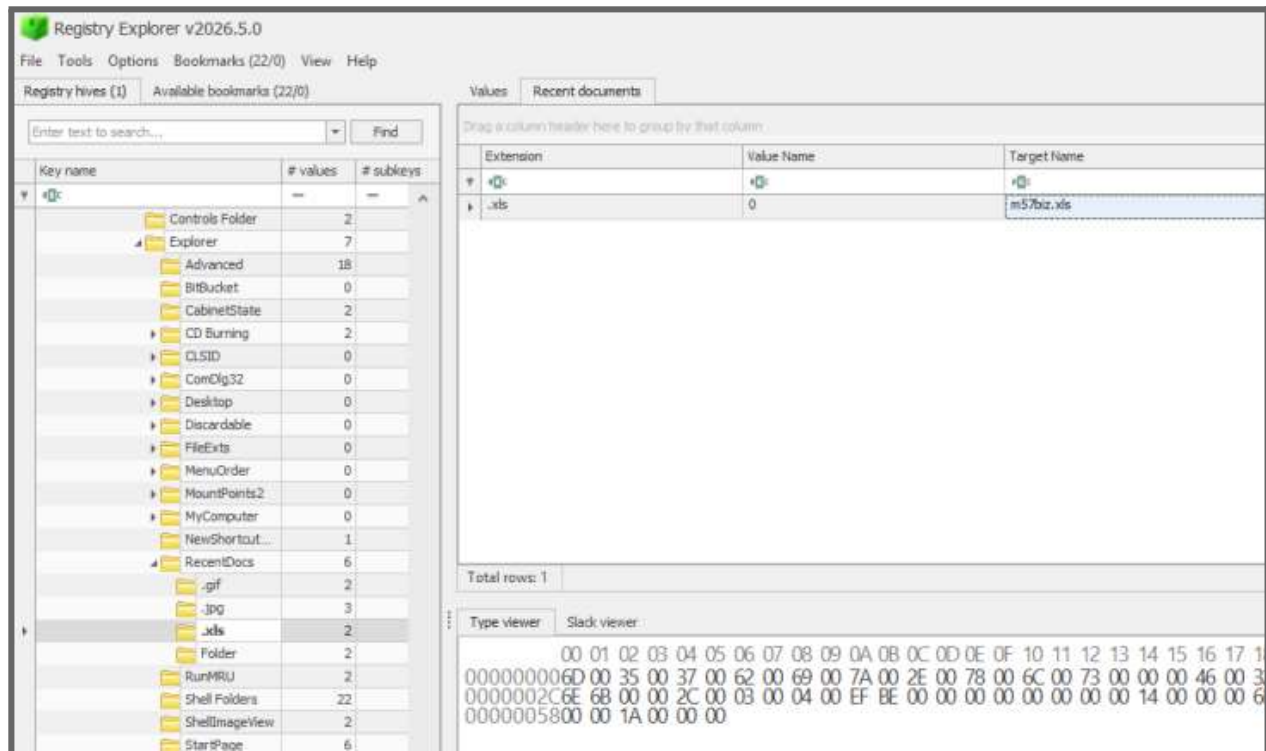
```

PECmd file references section highlighting the m57biz.xls entries

4.3 Registry Analysis (NTUSER.DAT)

Key 1 — RecentDocs: Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\.xls

Field	Value
Entry	m57biz.xls
MRU Position	0 (most recent)
Opened On	2008-07-20 01:28:04 UTC

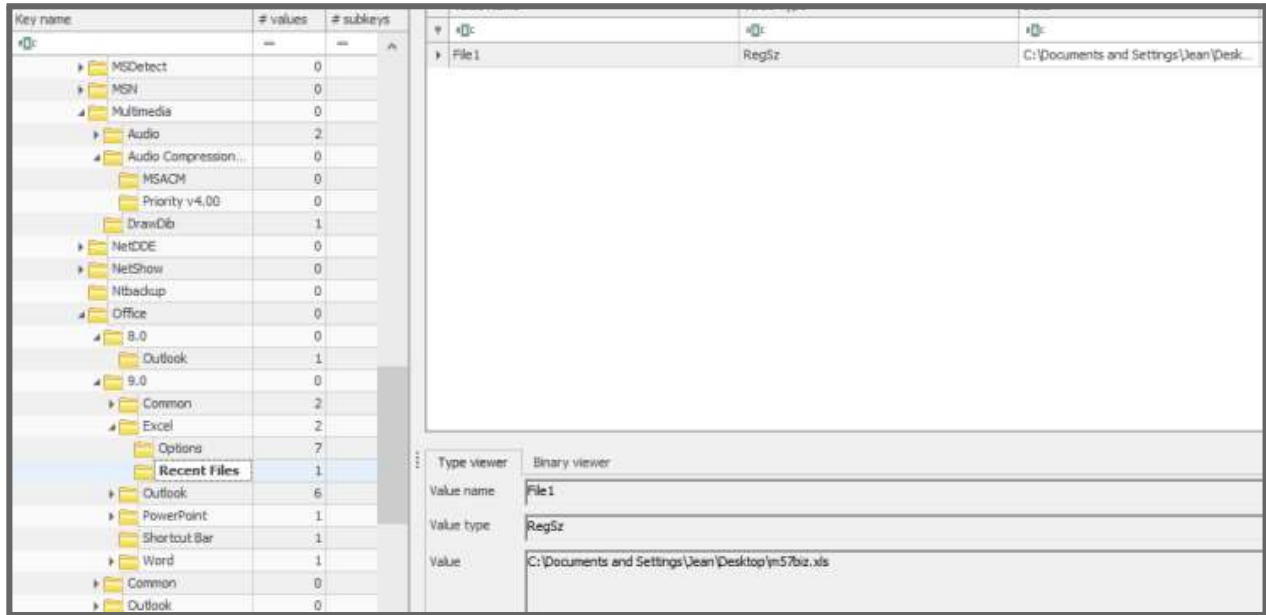


Registry Explorer RecentDocs .xls subkey showing m57biz.xls at MRU position 0

Key 2 — Excel Recent Files: Software\Microsoft\Office\9.0\Excel\Recent Files

Field	Value
Value Name	File1
Data	C:\Documents and Settings\Jean\Desktop\m57biz.xls
Last Write	2008-07-20 01:28:13 UTC

This last write (9 seconds after the email was sent) marks when Excel finalized its MRU after the file was closed.



Registry Explorer Excel Recent Files key showing File1 value

4.4 Artefact Correlation Table

Artefact	Timestamp (UTC)	User Action
Prefetch Last Run	01:27:40	Excel launched
Office Recent LNK Created	01:27:42	File opened in Excel
Prefetch Modified	01:27:50	Prefetch updated
Email Sent	01:28:00	Attachment sent

File MAC times updated	01:28:03	Outlook touched file
Windows Recent LNK Created	01:28:04	Shell shortcut written
RecentDocs Registry	01:28:04	MRU updated
Excel Recent Files Registry	01:28:13	Excel session finalized
Assessment: Three independent artefact categories converge within a 33-second window , conclusively proving Jean opened, handled, and sent the file. The artefacts are fully consistent with each other — no inconsistencies detected.		

5. Part C — Super Timeline Analysis (Days 79–80)

5.1 Timeline Generation

Autopsy's built-in Timeline tool was used. Filters applied in order:

- Date filter: July 20, 2008
- Time window: 01:20:00 – 01:30:45 UTC
- Keyword filter: m57biz

This reduced millions of raw events down to 10 highly relevant entries .

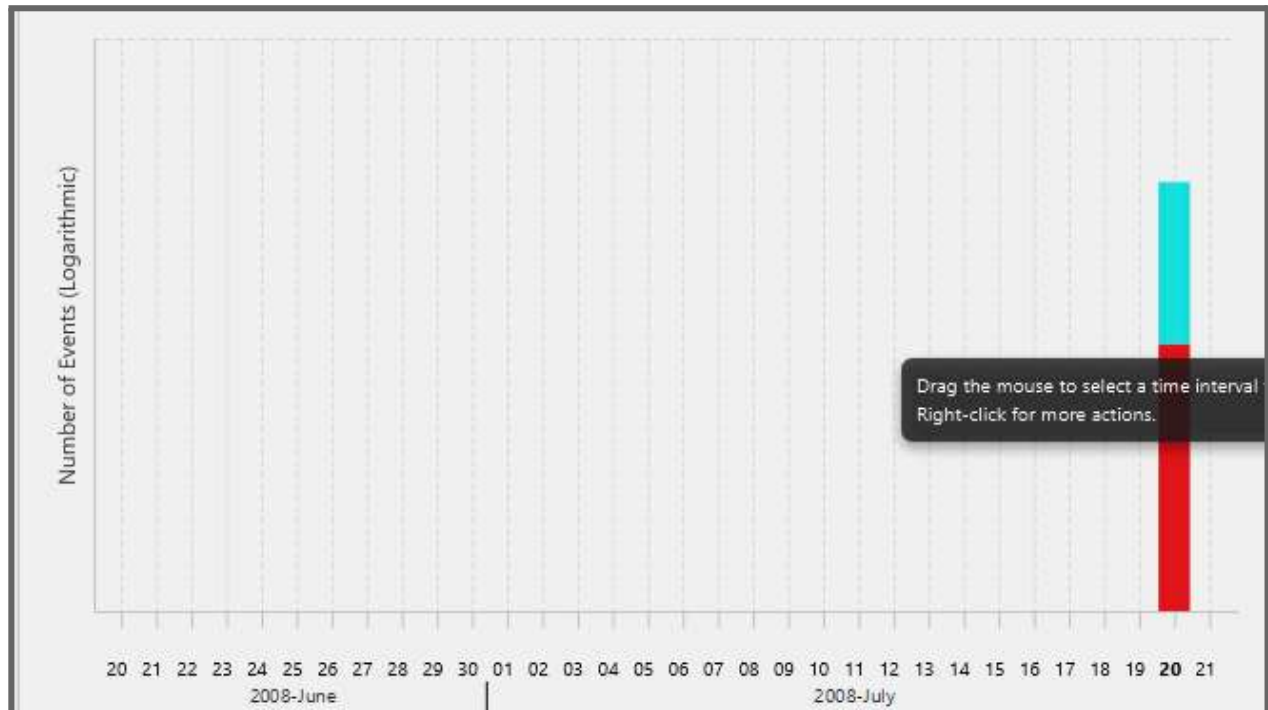
Date/Time	Event Type	Description	Tagged
2008-07-20 01:27:42	 Recent Docum...	C:\Documents and S ... esktop\m57biz.xls	
2008-07-20 01:27:42	 _B_	/Documents and Se ... Recent/m57biz.LNK	
2008-07-20 01:28:03	 _BM	/Documents and Set ... ook.pst/m57biz.xls	
2008-07-20 01:28:03	 A_BM	/Documents and Set ... esktop/m57biz.xls	
2008-07-20 01:28:04	 Recent Docum...	C:\Documents and S ... esktop\m57biz.xls	
2008-07-20 01:28:04	 AC_M	/Documents and Se ... Recent/m57biz.LNK	
2008-07-20 01:28:04	 _C_	/Documents and Set ... esktop/m57biz.xls	
2008-07-20 01:28:04	 ACBM	/Documents and Set ... Recent/m57biz.lnk	

Autopsy Timeline tool with filters applied, List view showing 8 results

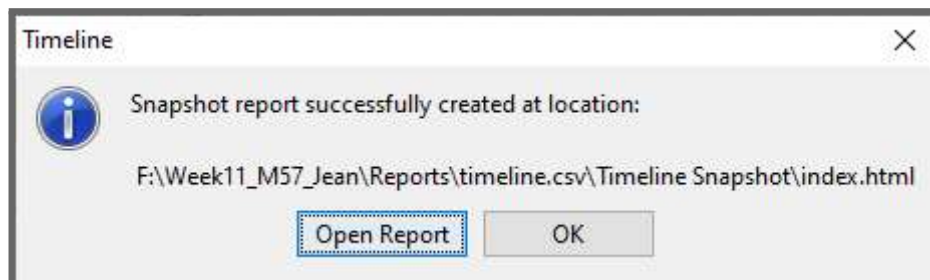
5.2 Filtered Timeline (8 Events)

#	Time (UTC)	Event Type	Description
1	01:27:42	Recent Docs	Desktop\m57biz.xls
2	01:27:42	B (Birth)	Office\Recent\m57biz.LNK
3	01:28:03	_BM	outlook.pst/m57biz.xls
4	01:28:03	A_BM	Desktop/m57biz.xls
5	01:28:04	Recent Docs	Desktop\m57biz.xls
6	01:28:04	AC_M	Office\Recent\m57biz.LNK
7	01:28:04	C (Changed)	Desktop/m57biz.xls
8	01:28:04	ACBM	Jean\Recent\m57biz.lnk

The 01:28:03 entry showing outlook.pst/m57biz.xls is especially significant — it ties the file directly to the outbound email in a single timeline row.



Counts View bar chart showing activity spike at minute 28



Snapshot Report export dialog saving timeline.csv to F:\Week11_M57_Jean\Reports\timeline.csv\Timeline Snapshot\index.html

5.3 Core 4-Event Sequence (Evidential Centrepiece)

#	Time (UTC)	Event	Artefact	Interpretation
1	01:27:40	Prefetch Run	EXCEL.EXE-1C75F8D 6.pf	Excel launched by Jean
2	01:27:42	LNK + RecentDocs	Office Recent + Registry	File opened in Excel
3	01:28:00	Email Sent	outlook.pst → Sent Items	Reply with attachment dispatched
4	01:28:03 -04	File / PST Modified	Desktop\m57biz.xls + PST	Outlook attached & saved file

6. Final Conclusion

Factual Findings

The investigation produced three independent layers of evidence that all corroborate each other. The email evidence (Part A) showed that the message Jean replied to displayed alison@m57.biz as the sender but the actual underlying delivery address embedded in the original headers was tuckgorge@gmail.com — an external Gmail account with no relationship to M57.biz. Jean's reply at 01:28:00 UTC therefore did not go to the President; it went to an external attacker. The host artefacts (Part B) — LNK files, the Excel prefetch, RecentDocs, and Excel's own MRU key — all independently recorded Jean opening [m57biz.xls](#) between 01:27:40 and 01:28:13 UTC. The super timeline (Part C) placed all of these events on one ordered view and showed Excel execution, file access, email transmission, and MFT updates clustered within ~33 seconds.

Analytical Conclusion

The evidence supports a finding of social engineering victimisation combined with negligence, rather than intentional malicious exfiltration. Jean was the target of a well-crafted spear phishing email that spoofed a trusted internal display name (alison@m57.biz) over an external Gmail address. However, Jean replied in just over 5 minutes without verifying the sender, without CC'ing anyone, and without questioning why an internal colleague would be sending from a Gmail account. As CFO handling SSNs, salaries, and payroll data, this represents a failure of professional due diligence. Jean did not intentionally leak the data, but she did send it — making her simultaneously a victim of the attack and a responsible party in the breach. The true threat actor is the operator of tuckgorge@gmail.com.

Sequence Summary

Time (UTC)	Event
01:22:45	Phishing email received (spoofed Alison)
01:27:40	Jean double-clicks file → Excel launches
01:27:42	LNK + RecentDocs created
01:28:00	Reply email with attachment sent to attacker
01:28:03	MFT timestamps updated by Outlook
01:28:04	Windows LNK + Registry finalized
01:28:13	Excel MRU last write — session closed

End of Report